Submission Details

Grade: **141.67 / 200**

M5: Exam 2 -- Covers modules 3 and 4

Jonathan Elder submitted Jun 26, 2022 at 10:49pm

ⓘ Correct answers are no longer available.

Score for this quiz: **141.67** out of 200
Submitted Jun 26, 2022 at 10:49pm
This attempt took 42 minutes.

Question 1 — 2 / 2 pts

$2 \in \{x|x+1<3\}$

- ○ True
- ◉ False

Partial

Question 2 — 1 / 2 pts

From the list below, select all values that are elements of the set:

$\{z|z^2-3z+2=0\}$

- ☑ 2
- ☐ -1
- ☑ 1
- ☐ 0
- ☐ -2
- ☐ 3
- ☑ -3

Question 3 — 2 / 2 pts

$5 \in \{\{1\},\{3\},\{5\},\{7\}\}$

- ○ True
- ◉ False

Question 4 — 2 / 2 pts

$3 \in \{-3,-1,1,3\}$

- ◉ True
- ○ False

Question 5

2 / 2 pts

Note that $\mathbb{Z}$ indicates the set of integers as described in the lecture videos.

$\mathbb{Z}^{-} \subset \mathbb{Z}$

- (selected) True
- False

Question 6

2 / 2 pts

$\{a, b, c\} \subset \{a, c, b\}$

- True
- (selected) False

Question 7

2 / 2 pts

Given any two sets X and Y, if $X \subseteq Y$ and $Y \subseteq X$ then $X = Y$.

- (selected) True
- False

Question 8

2 / 2 pts

$\{a, b, c, d, x\} \subset \{a, b, c, d\}$

- True
- (selected) False

Question 9

2 / 2 pts

$\{a, c, b\} \subseteq \{a, b, c\}$

- (selected) True
- False

Question 10

2 / 2 pts

Note that $\mathbb{Q}$ represents the set of **rational numbers** and $\mathbb{Z}$ represents the set of integers, as described in the lecture videos.

$\mathbb{Z} \subset \mathbb{Q}$

- (selected) True
- False

Question 11

8 / 8 pts

Given the following sets:

$X = \{1, 3, 5\}$

$Y = \{5, 1, 3\}$

Which of the following is the correct value for $R = X - Y$?

- $\emptyset$
- $\{-4, 2\}$
- $\{0\}$
- $\{-4, 2, 2\}$
- $\{1, 3, 5\}$

Question 12

8 / 8 pts

Given the following sets:

$X = \{1, 3, 5\}$

$Y = \{-2, 0, 1\}$

Which of the following is the set $P = X - Y$?

- $\{-2, 0, 1, 3, 5\}$
- $\{3, 5\}$
- $\{3, 3, 4\}$
- $\{3, 4\}$
- $\{-1, 0, 1, 2, 3, 5\}$
- $\{1, 3, 5\}$

Question 13

8 / 8 pts

Given the following sets:

$A = \{1, 2, 3\}$

$B = \{x | x \in \mathbb{Z} \wedge x < 2\}$

Select all elements from the following list that are in $A \cap B$.

- [ ] 4
- [ ] 2
- [x] 1
- [ ] 3

Incorrect

Question 14

0 / 8 pts

Given the following sets:

$R = \{x | x \text{ is odd} \wedge 0 < x < 9\}$

$S = \{x | x \in \mathbb{Z} \wedge x^2 < 2x + 3\}$

which of the following sets is equal to $R \cup S$?

Hint: If you're stuck, sketch a rough graph of $x^2$and $2x + 3$ so you can see what potential elements of S might be. Then, make a table with columns for x, $x^2$and $2x + 3$ and, based on your graph, list values of x where you think $x^2 < 2x + 3$ might hold. Complete your table and you should be able to list the elements of S. The elements of R should be easy to list.

- [x] $\{0, 1, 2, 3, 5, 7\}$
- [x] $\{1, 3, 5, 7\}$
- [ ] $\{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$
- [x] $\{0, 1, 3, 5, 7\}$

Question 15

8 / 8 pts

Given the following sets:

$A = \{2, 4, 6, 8\}$

$B = \{x | x \in \mathbb{Z} \wedge x < 4\}$

select all elements from the following list that are in $A \cap B$.

- [ ] 5
- [ ] 4
- [ ] 1
- [ ] 6
- [x] 2
- [ ] 3

Question 16

4 / 4 pts

Given the universal set $U = \{-3, -2, -1, 0, 1, 2, 3\}$which of the following is the complement of the set $S = \{1, 2, 3\}$?

- (•) $\{-3, -2, -1, 0\}$
- ( ) $\emptyset$
- ( ) $\{-3, -2, -1\}$

Question 17

4 / 4 pts

Given the universal set $U = \{x | x \in \mathbb{Z}^+ \wedge x^2 < 20\}$

Which of the following is the complement of $S = \{2, 3\}$?

- ( ) $\{0, 1, 2, 3, 4\}$
- (•) $\{1, 4\}$
- ( ) $\{-2, -3\}$
- ( ) $\emptyset$

Incorrect

Question 18

0 / 8 pts

Given arbitrary sets A and B, the following **set equality** is incorrect:

$(B - A) \cup A = A.$

Below are a list of potential **counterexamples**. Select all **counterexamples** that demonstrate **this identity is incorrect**.

- [x] $A = \{a, b\}, B = \{a, b, c\}$
- [ ] $A = \{a, b, c\}, B = \{a, b\}$
- [x] $A = \{a, b\}, B = \{a, b\}$
- [ ] $A = \{a\}, B = \{b\}$

Partial

Question 19

5.33 / 8 pts

In the following proof, select the set identity that is used to justify each step (note A and B are arbitrary sets and U is the universal set):

1. $\overline{A} \cup (A \cap B)$
2. $(\overline{A} \cup A) \cap (\overline{A} \cup B)$ [Select]
3. $U \cap (\overline{A} \cup B)$ [Select]
4. $\overline{A} \cup B$ [Select]

Answer 1:

Distributive

Answer 2:

Domination

Answer 3:

Identity

Question 20

4 / 4 pts

Given the sets:

$A = \{x | x \in \mathbb{Z}^+ \land 0 < x < 4\}$and

$B = \{a, b\}$

Which of the following is $A \times B$?

- ( ) $\emptyset$
- ( ) $\{a, b, 1, 2, 3\}$
- ( ) $\{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$
- (•) $\{(1, a), (2, a), (3, a), (1, b), (2, b), (3, b)\}$

Incorrect

Question 21

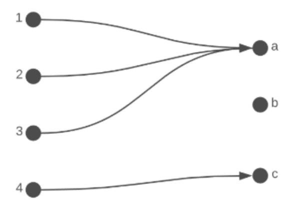
0 / 4 pts

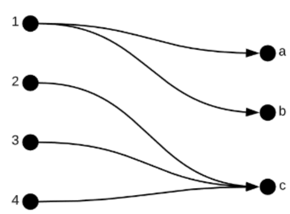
Given the sets:

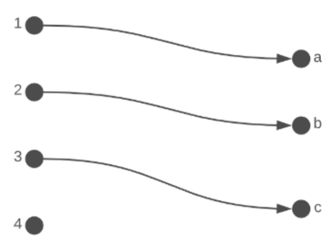
$A = \{1, 2, 3, 4\}$

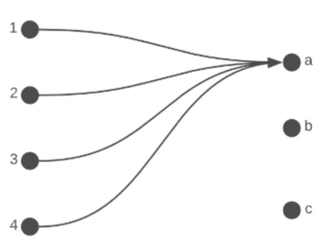
$B = \{a, b, c\}$

Select each of the following diagrams that define a function from A to B:









Question 22

4 / 4 pts

$f : \mathbb{Z} \to \mathbb{Z}$ where $f(x) = x^2 - 3$.

f is a function.

- (selected) True
- False

Question 23

4 / 4 pts

$g : \mathbb{R} \to \mathbb{R}$ where $g(x) = \pm\sqrt{x^2 + 4}$.

g is a function

- True
- (selected) False

Incorrect

Question 24

0 / 4 pts

What is the numerical value of $\lceil -1.8 \rceil$?

- -1.8
- -2
- 0
- -1
- -3
- (selected) 1.8

Question 25

4 / 4 pts

What is the numerical value of $\lceil 4 \rceil$?

- undefined
- 5
- -4
- 3
- 0
- (selected) 4

Incorrect

Question 26

0 / 4 pts

What is the numerical value of $\lfloor 1.2 \rfloor$?

- 1
- 2
- -1.2
- 0
- 1.2
- undefined

Question 27

4 / 4 pts

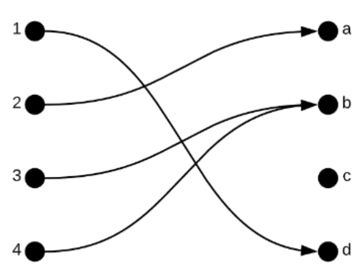
What is the numerical value of $\lfloor 4 \rfloor$?

- undefined
- 4
- 5
- 2
- 0
- 3

Question 28

4 / 4 pts

The function given by this diagram (as usual, domain on the left, co-domain on the right):



is a bijection (both one-to-one and onto).

- True
- False

Incorrect

Question 29

0 / 4 pts

Given the sets:

$A = \{-2, 0, 2\}$and

$B = \{1, 3\}$

it is possible to define an **onto function** $f : A \to B$.

- True
- False

Question 30

4 / 4 pts

Given the sets:

$A = \{-2, 0, 2\}$and

$B = \{1, 3\}$

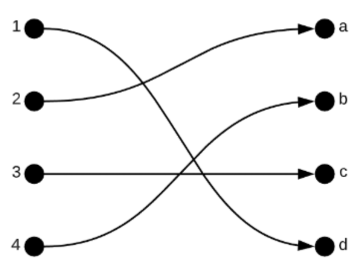
it is possible to define an **one-to-one function** $f : A \to B$.

- True
- False

Question 31

4 / 4 pts

The function given by this diagram (as usual, domain on the left, co-domain on the right):



is a bijection (both one-to-one and onto).

- True
- False

Question 32

4 / 4 pts

The relation R over the integers is defined as follows:

aRb iff $a + b = 0$.

Select each true statement from the list below.

- [ ] 3 R 1
- [x] -2 R 2
- [x] 0 R 0
- [ ] -1 R 3
- [x] 2 R -2

Incorrect

Question 33

0 / 8 pts

The relation R over the **positive integers** is defined as follows:

aRb iff $|a - b| < 2$

From the list below, select each property that holds for R.

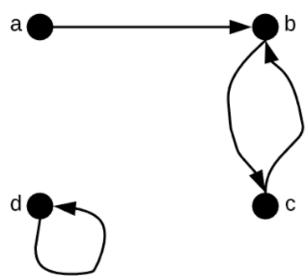
- [x] Anti-symmetric
- [ ] Symmetric
- [x] Reflexive
- [ ] Anti-reflexive
- [ ] Transitive

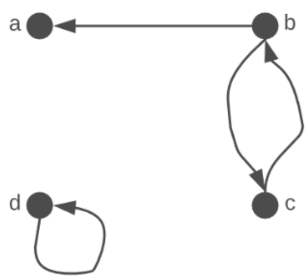
Question 34

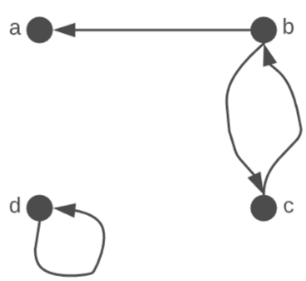
4 / 4 pts

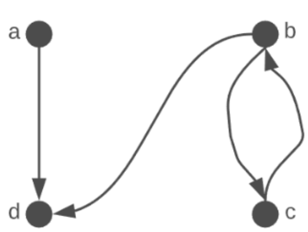
Which of the following digraphs represents the relation:

$P = \{(a,b),(c,b),(b,c),(d,d)\}$?





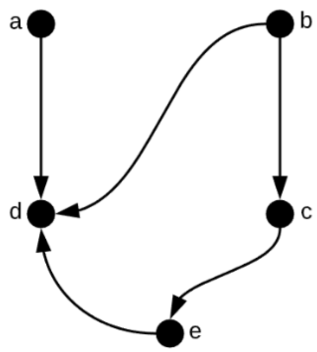




Question 35

8 / 8 pts

The following graph **contains a cycle.**



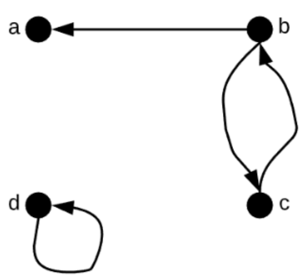
- True
- False

Partial

Question 36

4 / 8 pts

Given the digraph



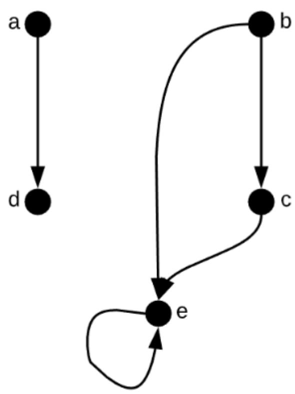
Select every **trail** in the list below

- [ ] <c,b,c,b,a>
- [x] <b,c,b>
- [x] <b,a>
- [x] <b,c,b,a>

Question 37

8 / 8 pts

Which of the following is the correct matrix representation of this graph:



Note: The rows correspond to vertices a, b, c, d and e, in that order. Similarly with the columns.

- $\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$

- $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

- $\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

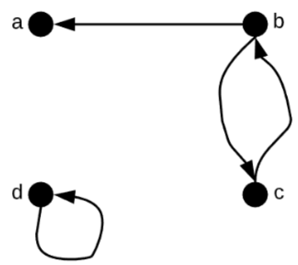
- $\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

Incorrect

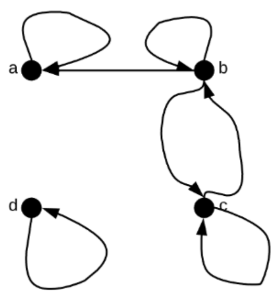
Question 38

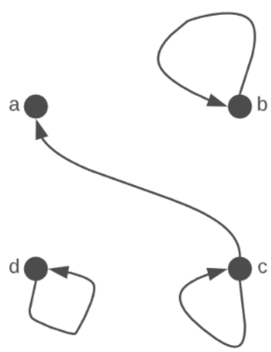
0 / 8 pts

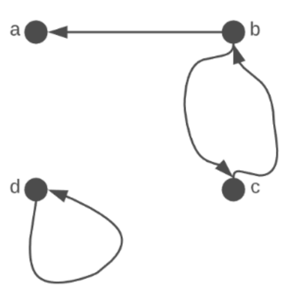
The following figure depicts the digraph G:



Which of the following is the correct figure for G^2?





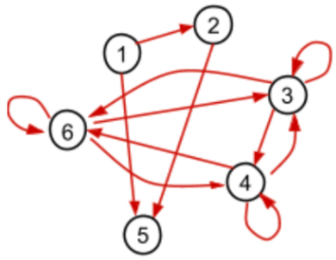


Partial

Question 39

5.33 / 8 pts

Suppose the transitive closure of the graph G is depicted in the following diagram:



From the following list, select facts that we can deduce about **the original graph G. Remember, only include facts that we can deduce with certainty.**

- [x] There must be an edge from 6 to 3
- [x] There must be an edge from vertex 1 to 2
- [ ] There must be a walk from 3 to 5
- [x] There must be no edge from vertex 1 to 6
- [x] There must be no walk from vertex 1 to 6
- [ ] There must be a path from vertex 1 to 6

Question 40

4 / 4 pts

Given the set of values

$S = \{4, 5, 12, 18, 22, 24, 31, 32, 41, 42, 43\}$

and the equivalence relation

$a \sim b$ iff a and b have the same remainder when divided by 3.

Select all of the elements below that are in the partition of S formed by this equivalence relation.

- [x] {5,32,41}
- [ ] {3,6,9,12}
- [x] {12,18,24,42}
- [ ] {4,5,12}
- [ ] {31,41}
- [x] {4,22,31,43}
- [ ] $\emptyset$

Question 41

4 / 4 pts

Given the set of values

$S = \{4, 5, 12, 18, 22, 24, 31, 32, 41, 42, 43\}$

and the equivalence relation

$a \sim b$ iff a and b have the same remainder when divided by 3.

How many equivalences classes does this relation have?

- 1
- 3
- 4
- 2
- 5
- 0

Question 42

4 / 4 pts

The sequence $R_i, i \in \mathbb{Z}^+$ is defined by the following recurrence relation:

$R_1 = 0$
$R_2 = 3$
$R_i = R_{i-1} + 2R_{i-2}$ for $i > 2$

What are the first 6 terms ($R_1, R_2, \ldots, R_6$) of this sequence?

- 0, 3, 10, 12, 14, 22
- 0, 3, 3, 6, 9, 15
- 0, 3, 3, 9, 15, 33
- 0, 1, 2, 3, 4, 5

Quiz Score: **141.67** out of 200